SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Debugging & Optimization Task

Course Code:	CSA0309	**Course Title:**	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	**Assessment:**	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	**Total Marks:**	100
CO2 Statement:	*Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.*		
Student Name:	G. Nithya Sree	**Reg. No.:**	192424103
Section / Batch:	3rd Year AIDS	**Date:**	22/07/2026

Code of Conduct

I G. Nithya Sree(192424103)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: G. Nithya Sree

Instructions

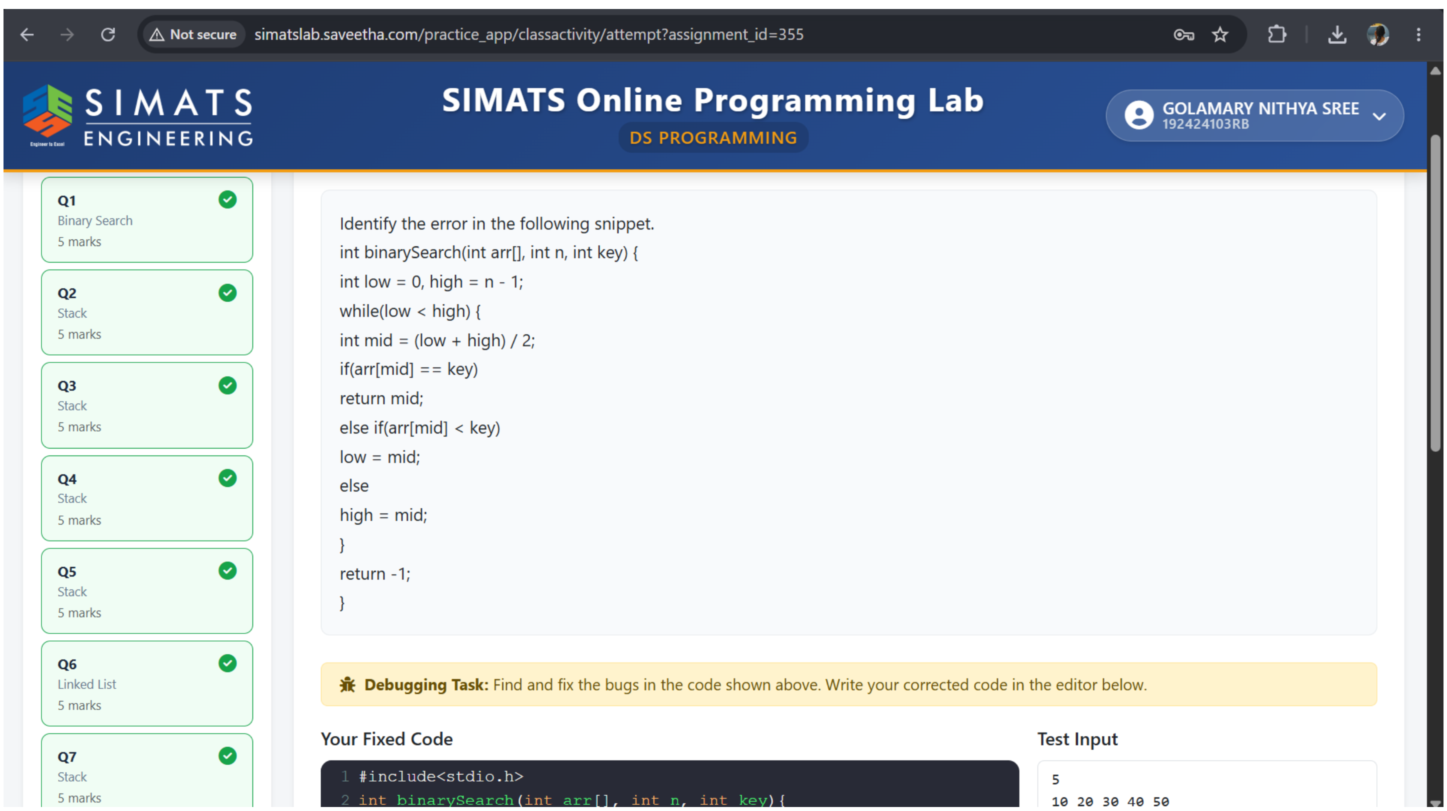
- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
		GRAND TOTAL:	**100 Marks**

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well-documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well-justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided

Faculty In-charge	Course Coordinator	Program Director
Signature: ____________	Signature: ____________	Signature: ____________
Date: ________________	Date: ________________	Date: ________________



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Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List Data Structures
5 marks

```
if(top == 0) {
printf("Underflow");
return -1;
}
return stack[top--];
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include<stdio.h>
int stack[100];
int top=-1;
int pop(){
    if(top==-1){
        printf("UnderFlow\n");
        return -1;
    }
    return stack[top--];
}
int main(){
    pop();
    return 0;
```

Test Input

stdin input (optional)

Output

```
UnderFlow
```

32° ENG IN 11:32 05-08-2026

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Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List Data Structures
5 marks

```
void init() {
top = 0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include<stdio.h>
int stack[5];
int top;
void init(){
    top=-1;
}
int main(){
    init();
    return 0;
}
```

Test Input

stdin input (optional)

Output

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QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q4 **Stack** 5 marks

Identify the errors in the following snippet.

```
if(ch == ')') {
while(stack[top] != ')') {
postfix[j++] = pop();
}
pop();
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
if(ch==')'){
    while(stack[top]!=')'){
        postfix[j++]=pop();
    }
    pop();
}
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack 5 marks
- Q6 Linked List 5 marks

Q5 **Stack** 5 marks

In balancing Bracket, you identify the error.
if((ch == ']' && stack[top] == '[') ||
(ch == '}' && stack[top] == '{')) {
pop();
} else {
return 0;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include<stdio.h>
#include<string.h>
char stack[100];
int top=-1;
void push(char x){
    stack[++top]=x;
```

Test Input

stdin input (optional)

Output

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- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack 5 marks
- Q6 Linked List 5 marks
- Q7 Stack 5 marks

What is the issue?
temp = head
while temp.next != NULL:
print(temp.data)
temp = temp.next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include<stdio.h>
#include<stdlib.h>
struct Node{
    int data;
    struct Node* next;
};

void display(struct Node* head){
    struct Node* temp = head;
    while(temp != NULL){
        printf("%d", temp->data);
        temp = temp->next;
    }
```

Test Input

stdin input (optional)

Output

```
1020
```

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- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack 5 marks
- Q6 Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List Data Structures 5 marks
- Q9 Linked List

else:
return stack[top--]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include<stdio.h>
int top=-1;
void POP(int stack[]){
    if(top==-1){
        printf("UnderFlow\n");
    }
    else{
        printf("%d\n", stack[top--]);
    }
}
int main(){
    int stack[100];
    POP(stack);
    return 0;
}
```

Test Input

stdin input (optional)

Output

```
UnderFlow
```

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Submitted

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Q5 Stack 5 marks
Q6 Linked List 5 marks
Q7 Stack 5 marks
Q8 Linked List Data Structures 5 marks
Q9 Linked List 5 marks
Q10 Queue 5 marks

8/10 answered

```c
struct Node{
    int data;
    struct Node* next;
};
struct Node* reverse_list(struct Node* head){
    struct Node* prev=NULL;
    struct Node* current=head;
    struct Node* next_node=NULL;
    while(current!=NULL){
        next_node= current->next;
        current->next=prev;
        prev=current;
        current=next_node;
    }
    return prev;
}
int main(){
    struct Node* head=(struct Node*)malloc(sizeof(struct Node));
    struct Node* second=(struct Node*)malloc(sizeof(struct Node));
    head->data=-1;
    head->next=second;
    second->data=2;
    second->next=NULL;
    head=reverse_list(head);
    struct Node* temp=head;
    while(temp!= NULL){
```

Output

```
2-1
```

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Q4 Stack 5 marks
Q5 Stack 5 marks
Q6 Linked List 5 marks
Q7 Stack 5 marks
Q8 Linked List Data Structures 5 marks
Q9 Linked List 5 marks
Q10 Queue 5 marks

```c
#include<stdlib.h>
struct Node{
    int data;
    struct Node* next;
};
int main(){
    struct Node* head=(struct Node*)malloc(sizeof(struct Node));
    head->data=20;
    head->next=NULL;
    struct Node* new=(struct Node*)malloc(sizeof(struct Node));
    new->data=10;
    new->next=head;
    head=new;
    struct Node* temp=head;
    while(temp!= NULL){
        printf("%d", temp->data);
        temp=temp->next;
    }
    return 0;
}
```

Submitted

Output

```
1020
```

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5 marks
Q7 Stack 5 marks
Q8 Linked List Data Structures 5 marks
Q9 Linked List 5 marks
Q10 Queue 5 marks

10/10 answered

```c
    }
    else{
        int item=Q[front];
        if(front==rear){
            front=-1;
            rear=-1;
        }
        else{
            front++;
        }
        return item;
    }
}
int main(){
    int Q[100]={10, 20, 30};
    DEQUEUE(Q);
    return 0;
}
```

Submitted

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Q1 Stack Notations - Postfix Expression 5 marks
Q2 Circular Queue 5 marks
Q3 Binary Search 5 marks
Q4 Queue 5 marks
Q5 Queue 5 marks
Q6 Doubly Linked List 5 marks
Q7 Stack

Find the bug in the evaluation of a Postfix expression.
while (expression[i] != '\0') {
if (isdigit(expression[i])) push(expression[i] - '0');
else {
int op1 = pop();
int op2 = pop();
// perform operation: result = op1 + op2;
push(result);
}
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#include <ctype.h>

#define MAX 100

int stack[MAX];
int top = -1;
```

Test Input

stdin input (optional)

Output

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Stack Notations - Postfix Expression 5 marks
Q2 Circular Queue 5 marks
Q3 Binary Search 5 marks
Q4 Queue 5 marks
Q5 Queue 5 marks
Q6 Doubly Linked List 5 marks
Q7 Stack 5 marks

Identify the bug in this Circular Queue dequeue logic.
int dequeue() {
if (front == -1) return -1;
int data = queue[front];
front = (front + 1) % MAX;
return data;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#define MAX 5

int queue[MAX];
int front = -1, rear = -1;

void enqueue(int data)
{
    if ((rear + 1) % MAX == front)
    {
        printf("Queue Overflow\n");
        return;
```

Test Input

stdin input (optional)

Output

```
Queue: 10 20 30 40
Deleted Element = 10
Deleted Element = 20
Queue: 30 40
Queue: 30 40 50 60
```

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Stack Notations - Postfix Expression 5 marks
Q2 Circular Queue 5 marks
Q3 Binary Search 5 marks
Q4 Queue 5 marks
Q5 Queue 5 marks
Q6 Doubly Linked List 5 marks
Q7 Stack 5 marks

Identify the bug in this Recursive Binary Search.
int binarySearch(int arr[], int l, int r, int x) {
int mid = l + (r - l) / 2;
if (arr[mid] == x) return mid;
if (arr[mid] > x) return binarySearch(arr, l, mid, x);
return binarySearch(arr, mid, r, x);
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>

int binarySearch(int arr[], int l, int r, int x)
{
    if (l > r)
        return -1;

    int mid = l + (r - l) / 2;

    if (arr[mid] == x)
        return mid;
```

Test Input

stdin input (optional)

Output

```
Element found at index 5
```

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Stack Notations - Postfix Expression
5 marks

Q2 Circular Queue 5 marks

Q3 Binary Search 5 marks

Q4 Queue 5 marks

Q5 Queue 5 marks

Q6 Doubly Linked List 5 marks

Q7 Stack 5 marks

Find the error in this Deque insertFront operation.
void insertFront(int data) {
if (front == 0) front = MAX - 1;
else front = front - 1;
deque[front] = data;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>

#define MAX 5

int deque[MAX];
int front = -1, rear = -1;

void insertFront(int data)
{
    // Check Overflow
    if ((front == 0 && rear == MAX - 1) || (front == rear + 1))
    {
        printf("Deque Overflow\n");
```

Test Input

stdin input (optional)

Output

30 20 10

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

Back

QUESTIONS

Q1 Binary Search 5 marks

Q2 Stack 5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Stack 5 marks

Q6 Linked List

Q5 **Stack** 5 marks

In balancing Bracket, you identify the error.
if((ch == ']' && stack[top] == '[') ||
(ch == '}' && stack[top] == '{')) {
pop();
} else {
return 0;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include<stdio.h>
#include<string.h>
char stack[100];
int top=-1;
void push(char x){
```

Test Input

stdin input (optional)

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

Back

QUESTIONS

Q1 Binary Search 5 marks

Q2 Stack 5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Stack 5 marks

Q6 Linked List

Q6 **Linked List** 5 marks

What is the issue?
temp = head
while temp.next != NULL:
print(temp.data)
temp = temp.next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include<stdio.h>
#include<stdlib.h>
struct Node{
    int data;
    struct Node* next;
};

void display(struct Node* head){
```

Test Input

stdin input (optional)

Output

1020

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QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack 5 marks
- Q6 Linked List 5 marks
- Q7

Q7 Stack

5 marks

Point out the error.
POP(stack):
if top == 0:
print("Underflow")
else:
return stack[top--]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include<stdio.h>
int top=-1;
void POP(int stack[]){
    if(top==-1){
        printf("UnderFlow\n");
    }
    else{
        printf("%d\n", stack[top--]);
    }
}
```

Test Input

stdin input (optional)

Output

```
UnderFlow
```

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QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack 5 marks
- Q6 Linked List 5 marks

Q8 Linked List Data Structures

5 marks

Identify the errors in the following code
while current != NULL:
current.next = prev
prev = current
current = current.next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include<stdio.h>
#include<stdlib.h>
struct Node{
    int data;
    struct Node* next;
};
struct Node* reverse_list(struct Node* head){
    struct Node* prev=NULL;
    struct Node* current=head;
    struct Node* next_node=NULL;
```

Test Input

stdin input (optional)

Output

```
2-1
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack 5 marks
- Q6 Linked List

Q9 Linked List

5 marks

What is the error in the below code?
new.next = head
head = new

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include<stdio.h>
#include<stdlib.h>
struct Node{
    int data;
    struct Node* next;
};
int main(){
    struct Node* head=(struct Node*)malloc(sizeof(struct Node));
    head->data=20;
    head->next=NULL;
    struct Node* new=(struct Node*)malloc(sizeof(struct Node));
```

Test Input

stdin input (optional)

Output

```
1020
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

Back

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List

Q10 **Queue** 5 marks

What is missing in the below pseudocode?
DEQUEUE(Q):
if front == -1:
print("Underflow")
else:
return Q[front++]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include<stdio.h>
int front=0;
int rear=2;
int DEQUEUE(int Q[]){
    if(front==-1){
        printf("UnderFlow\n");
```

Test Input

stdin input (optional)

Output